

Arithmetic Sequences and Series

ANSWER KEY



Arithmetic sequences

- 1 Consider the sequence 7, 11, 15, 19, ...
- a** State the common difference. $d = 4$
- b** Write a formula for the n th term. $a_n = 4n + 3$
- c** Find a_{20} . $a_{20} = 83$
- 2 In an arithmetic sequence, $a_3 = 10$ and $a_8 = 25$.
- a** Find the common difference and the first term. $5d = 15$ gives $d = 3$; $a_1 = 10 - 2(3) = 4$.
- b** Write a formula for a_n . $a_n = 4 + 3(n - 1) = 3n + 1$.

Arithmetic series

- 3 Find the sum of the first 30 terms of the series $5 + 9 + 13 + \dots$
 $a_{30} = 5 + 29(4) = 121$, so $S_{30} = \frac{30(5 + 121)}{2} = 1890$.
- 4 Consider the multiples of 6 strictly between 100 and 500.
- a** How many are there? First 102, last 498: $n = \frac{498 - 102}{6} + 1 = 67$.
- b** Find their sum. $S = \frac{67(102 + 498)}{2} = 67 \times 300 = 20,100$.
- 5 A theatre has 18 seats in the first row, and each subsequent row has 3 more seats than the row before. How many seats are there in total in 25 rows?
 $a_{25} = 18 + 24(3) = 90$, so $S_{25} = \frac{25(18 + 90)}{2} = 1350$ seats.